Project Demo: Existing Implementation and Proposed Work (Differentiation)

NeuroSonic project · Allwin Raj · Vellore Institute of Technology · 3 November 2026

Project Demonstration: Existing Implementation and Proposed Work

The demonstration separates three things: (A) what the base paper provided, (B) what this project implemented and can demonstrate now, and (C) what is proposed but not yet implemented or not yet physically verified.

Table 1. (A) Base work received (not this project's contribution)

Item	State when received
Cross-modal detector (NIR + depth), exp72 checkpoint	scripts broken with current libraries
Reported accuracy mAP@0.5 0.669	training plot only; not reproducible with published data
Published dataset	a different dataset (other classes; none of the run's validation frames)
FPGA, RL compression, Android app, SDK	not used (SDK only for the colour-map finding)

Table 2. (B) Implemented in this project and demonstrable now

#	Contribution	Verification level
B1	Faithful standalone WOAD inference (bit-identical)	automated tests
B2	Dataset audit and evaluation (person AP@0.5 0.218 / 0.173)	measured
B3	Alert engine and input-quality gate (replay of recorded NIR + depth)	automated tests
B4	NeuroSonic server, API (28 endpoints), SQLite text history, accessible phone UI	automated + browser tests
B5	Detector benchmark; RF-DETR Nano FP16 on GPU, SSDLite CPU fall-back	measured; tests
B6	Camera-motion-compensated tracking; entering / crossing events	tests; composited clip
B7	Piper neural voice; priority speech queue	browser tests (laptop)
B8	Vosk offline voice commands	synthetic speech; browser test with fake mic
B9	Offline Vellore map, search, A* routing, spoken turns, re-routing	tests; simulated walk
B10	HTTPS with local certificate authority; phone setup page; release start/stop scripts	tests; launch verified
B11	Optional opt-in cloud description with wording filter	mock-provider tests

Table 3. (C) Proposed work, not yet implemented or not yet physically verified

#	Proposal	Reason
C1	iPhone and field validation (23 scenarios × 5 trials)	nothing verified physically yet
C2	Monocular depth for "approaching" alerts	no distance from an RGB camera today
C3	Ground-hazard data: steps, kerbs, potholes, poles	outside the 23 COCO classes
C4	Native app with on-device inference	removes the laptop dependency
C5	Better recognition of spoken place names	10 of 24 synthetic utterances correct first
C6	User study with visually impaired participants (with ethics approval)	thresholds and wording are engineering defaults

Demonstration script (about 10 minutes)

1. **Base paper (2 min).** Run the re-implemented WOAD detector on a recorded NIR + depth sequence (Settings → Research tools → WOAD replay); show the equivalence test result and the evaluation table.
2. **Launch (1 min).** `START_NEUROSONIC.ps1`; show the printed detector (RF-DETR, cuda, fp16) and the iPhone address.
3. **Walking-path awareness (3 min).** On the phone (or the laptop browser with the demo clip), press Start; show "appears clear", "Caution", "Stop", and the event "A person is moving into your path from the left"; pan the camera over a static scene to show that no event is raised.
4. **Voice and navigation (2 min).** Say "navigate to CMC Hospital"; confirm; show the offline route and spoken turn instruction; show that it works with the internet disconnected.
5. **Honesty and limits (2 min).** Show the "How it works" screen; state what is not yet verified.